



Application of Bayesian network and artificial intelligence to reduce accident/incident rates in oil & gas companies

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ABSTRACT

Process safety management (PSM) is a framework that demonstrates a company's commitment to process safety, a better understanding of hazards and risks, a comprehensive assessment and management of risks, and enhanced learning from experience to improve overall safety and operational performance. Companies often use an incident data reporting system to execute PSM. While companies keep incident data in thousands of reports, rarely do they glean full value in learning from these to prevent and reduce future incidents. To overcome this challenge, this research applied machine learning and keyword analysis to label and classify 8199 incident reports from an oil and gas company into nine groups identified in the latest version of PSM guidelines published by the Center for Chemical Process Safety (CCPS). To converge on an optimal solution, two different Bayesian network techniques (Tabu and hill climbing) were applied. Both methods resulted in the same map, showing that the Total Number of Incidents has the maximum dependency (50%) on Asset Integrity & Reliability; this means focusing resources on this aspect could reduce the total number of incidents by half. Cross correlation analysis (CCA) was also applied, which validated and confirmed this result. This analysis identifies which measures enhance the company's safety management strategy to reduce these latent causes, but also supports critical thinking, enhanced communication, and learning culture to improve organizational safety.

1. Introduction

Process safety management (PSM) and risk management research has grown exponentially since the 1960s; yet, academic researchers dominate this work and rarely collaborate with industrial partners, examine data sources, or consider applications (Amin et al., 2019). Unfortunately, major industrial accidents and fatalities persist. Production loss, absenteeism, medical costs, and compensation can total up to 4% of the annual global gross domestic product (Takala et al., 2014). We argue that to have the greatest accessibility, uptake, and impact on process safety practice, academic researchers, such as ourselves, must examine industrial partners' data, frame our work using their PSM frameworks, and provide an analysis that lends itself to actionable recommendations.

The oil and gas industry has expanded greatly in recent years; in the United States alone, the oil and gas extraction industry nearly doubled in size from 2003 to 2013 and increased the number of drilling rigs by 71%. During this period of growth, the fatality rate was seven times higher than all other industries combined in the United States. When

compared to other industries in the mining sector, 74% of the fatalities in 2015 occurred in the oil and gas industry vs. only 26% in other mining industries. From 2003 to 2015, a total of 1189 deaths were confirmed in the US oil and gas extraction industry, of which at least 275 were related to process safety (Smith, 2018). From 2015 to 2017, a further 161 fatalities occurred of which 55 were related to process safety (Centers for Disease Control and Prevention (CDC), 2020).

As the oil and gas industry continues to grow, so does the need to focus on the safety of people and the environment. A massive portion of the risk found in this industry involves process safety and, as such, PSM seeks to manage hazards associated with process industries, consequently reducing risks involving the release of hazardous chemicals (OSHA, 2000). Many process safety incidents result from minor errors that compound to form a major disaster (Ness, 2015). Examples include the explosion of a natural gas condensate tank at Warffum, Netherlands that resulted in two fatalities and the BP America refinery explosion in Texas, USA that resulted in 15 deaths and 180 injuries (Knegtering & Pasman, 2009). Both examples involved hazardous chemical substances, included multiple flaws that compounded into a larger

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accident, and could have been prevented with proper safety management (Knegtering & Pasman, 2009; Hudson et al., 1991). Many more incidents over the past few decades also demonstrate how effective PSM could have prevented major losses (Khan et al., 2016). Khan et al. (2016) state that, for PSM to be effective, it must be dynamic to ensure continuous safety improvement, incorporate new findings, and develop a strong safety culture. At the same time, failure to implement adequate process management systems has been identified as a key contributor to incidents such as the Deepwater Horizon explosion in 2010. Organizations must uphold a certain standard when dealing with process safety to achieve process safety excellence (Amyotte et al., 2007).

In North America, two key organizations provide PSM standards: the Occupational Safety and Health Administration (OSHA), as part of the United States Department of Labor, and the Center for Chemical Process Safety (CCPS), which is part of the American Institute of Chemical Engineers (AIChE) and provides cross-sector industrial initiatives (CCPS, 2011). To deal with process safety, the OSHA has defined 14 elements of PSM that companies are required to meet (OSHA, 2000). CCPS has provided 20 PSM elements for companies to consider; these elements include and exceed the requirements set by the OSHA and provide many benefits towards future development and risk reduction (CCPS, 2011).

Unlike studies that attempt to classify incident reports into categories decided by individuals (Mazaheri et al., 2015), this research seeks to classify incident reports into the PSM elements defined by CCPS. The PSM elements defined by CCPS were selected due to their stringent requirements that encompass the mandatory regulations set by the government. Classifying incident reports into defined elements of PSM has twofold benefits: the classification itself can be justified (unlike the results of surveys, which can sometimes be considered arbitrary) and companies can be assured knowing that the recommendations provided by such research meet stringent requirements that surpass government regulations.

Fig. 1 illustrates the 20 PSM elements defined by CCPS built upon four foundational blocks: *Commit to Process Safety*, *Understand Hazards and Risk*, *Manage Risk*, and *Learn from Experience* (CCPS, 2011). The AIChE defines process safety as an organizational commitment. An organization that focusses on developing a culture of excellence in process safety will see improvement in many areas, such as leadership,

management, and workforce. *Commit to Process Safety* includes five PSM elements and promotes the active involvement of personnel at all levels of the organization. *Understand Hazards and Risk* includes two PSM elements and involves using hazard and risk information in an effective manner from early stages of project development towards long-term success. *Manage Risk* has nine dedicated PSM elements and is defined as an ongoing commitment to manage processes that pose risk to ensure the risk remains tolerable and to remain prepared for and manage incidents that may occur. Finally, the fourth foundational block is *Learn from Experience*, and involves monitoring incidents, both within and outside the organization, and to use lessons learned to prevent incidents from recurring.

Many industrial organizations analyze their incident reports—such as chemical facilities, rail transport systems, or civil sectors—as part of their safety management systems (Schaaf & Kanse, 2004) to determine root causes and share lessons learned to prevent similar incidents from reoccurring in the future. This is also the role of the U.S. Chemical Safety and Hazard Investigation Board (U.S. CSB, 2014). Yet, technological, organizational, and human factors are often interacting (Jitwasinkul et al., 2016) in complex, multi-directional, and uncertain ways, which can be difficult to disentangle in the aggregate.

Bayesian network (BN) analysis can display both horizontal and vertical dependencies, data and knowledge uncertainty, and practical applications (Amin et al., 2019). Recognizing this, our research develops a unique analytical approach using classification of the incident data by keyword analysis and developing the most probable network by the application of Bayesian theory and validating the result by cross-correlation analysis (CCA). The data analyzed in this research come from the incident database of a large oil and gas company. In our previous work, these incident reports were analyzed with the goal of classifying incidents and providing outputs (Kurian et al., 2020a) that could be used by companies to reduce risk on work sites. The purpose of the present article is to extend this research by focusing on incident reports involving different aspects of PSM, analyzing their relative contribution to the overall risk, and, as a result, reducing the number of incidents occurring on site.

Initially, we searched for the most important clusters of PSM elements that can later be present in a probabilistic causality model such as a BN. This step determined nine PSM elements and later grouped all

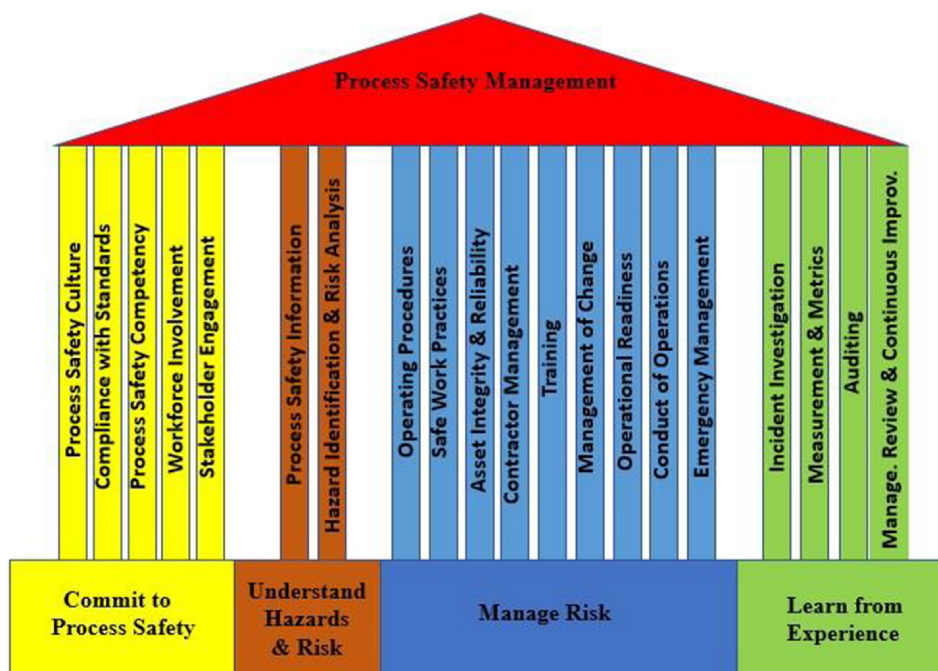


Fig. 1. Risk-based process safety management elements (CCPS, 2011).

incidents into these clusters. To develop the link and interconnectivity between these clusters, we had to overcome the challenge of dealing with a high-dimensional dataset (8199 variables). A major issue in computational fields such as biology or chemical processes is the use of high-dimensional datasets to accurately study the network architecture of the variables (Friedman et al., 1999; Tsamardinos et al., 2006). Several studies have been conducted on the systematic and computationally intensive comparisons between different techniques on synthetic experimental data, and infer that Bayesian inference is preferable when dealing with a high-dimensional dataset (Zou et al., 2009; Jansen et al., 2003). For this very reason, the BN approach has increasingly been used in various disciplines in recent years, including risk analysis for construction projects (Wu et al., 2015), medical diagnosis (Heckerman & Mamdani, 1995), and even safety control and accident analysis (Jitwasinkul et al., 2016).

A BN is an analytical form that depicts a cumulative probability distribution among a set of random variables. The present research goes one step further and uses two different BN techniques—Tabu and hill climbing (HC)—to provide an optimal solution (Jiang et al., 2013a). Another aspect of using these techniques is analyzing the network that maximizes the score function showing how the network optimally fits the data. These artificial intelligence (AI) and machine learning (ML) techniques delivered a quantitative framework to analyze the incident dataset from an oil and gas company. The algorithm was able to determine the importance of each contributing factor, prioritize them, and map the way they are linked (Mazaheri et al., 2015). Ultimately, this approach aims to reduce the number of incidents by providing recommendations to address these priority factors.

2. Methodology

2.1. Data collection, machine learning, and keywords analysis

This study builds on research that used machine learning and keyword analysis to categorize and analyze incident reports (Kurian et al., 2020b). The objective was to use the elements of PSM defined by CCPS to classify incident reports. The provincial government has imposed several criteria for incident reports; they are required to contain the location of the incident, time and date of occurrence, name of the employer involved, contact information of the site contact, and a general description of the incident (Government of Alberta, 2019). This research analyzed incident reports from 2014 (15,674 reported incidents) and 2015 (16,177 reported incidents). Several incident reports were incomplete, making it difficult to assemble the information necessary for classification. The company providing the incident database used in this research had already completed preliminary investigations for many of their incident reports, which included root cause analysis. The first step was to narrow the scope of incidents being analyzed to the incident reports with a completed root cause analysis. This made it possible to remove the incident reports lacking the information necessary for classification.

The second step was to determine which CCPS elements of PSM should be used to classify the incident reports. Elements were selected from each of the four foundational blocks: *Compliance with Standards* from *Commit to Process Safety*; *Process Safety Information* and *Hazard Identification & Risk Analysis* from *Understand Hazards and Risk*; *Operating Procedures*, *Safe Work Practices*, *Asset Integrity & Reliability*, *Contractor Management*, and *Training* from *Manage Risk*; and *Management Review & Continuous Improvement* from *Learn from Experience*. The primary tool for classification was keyword analysis and, as such, these PSM elements were selected based on the information typically found in the incident reports. In other words, the elements not selected would introduce interpretation bias given the lack of information in the database analyses. This lack of information is a product of the standards of the system used. For example, it would be difficult to classify a simple slip/trip/fall incident as a violation of a

specific PSM element without further details, e.g., such an incident could be classified as a violation of several elements from both the *Understand Hazards and Risk* and the *Manage Risk* foundational blocks.

Next, the incidents were classified according to the PSM elements selected in the previous step. The incidents were recorded in a table including event date, event description, root cause, PSM elements, and total as headings. The first three columns were included in the incident database while the PSM elements were classified using keyword analysis. If an incident matched with a PSM element, a value of 1 was assigned to the incident under the column of the corresponding element. Conversely, a value of 0 was assigned to elements that did not match the incident. An incident could match more than one element and, as such, totals were calculated for each row as the sum of the element values in the specified row. Finally, the 8199 incidents classified were sorted by date to deliver the total number of incidents per day for each PSM element. These steps were done by the newest Python version, 3.8.1.

2.2. Learning Bayesian network structure

The next step after cluster analysis was developing and optimizing the cluster interconnected network architecture. Learning techniques usually solve the optimization issue by applying accepted heuristic search procedures. When the search space is very large, the application of such techniques, which tend to invest resources analyzing contenders, cannot be logical. This issue becomes challenging for huge datasets, either in terms of the number of instances or number of attributes (Friedman et al., 1999), similar to the dataset obtained by this research. Therefore, this research utilized the BN procedure to tackle this issue (Sattari et al., 2020). Besides presenting causal relationships, BNs can handle missing data, avoid overfitting, and combine data with expert knowledge (Uusitalo, 2007).

BNs are often perceived as an optimization approach where the function of data processing is to identify the network that increases a mathematically instigated score. Bayesian learning networks are used to develop the most probable reaction network based on the data. The BN can represent the quantitative strength of the connections between clusters found in the previous steps. A BN is a joint probability distribution including a series of random variables (V). This network is characterized by a pair $B = (G, \Phi)$, in which G represents a directed acyclic graph with nodes $X_1, X_2, \dots, X_n$; Φ represents random variables, V ; and edges represent the direct dependencies between these variables. The graph, G , encodes independence assumptions by which each variable, X_i , is independent of its non-descendants given its parents in G (Lee & Abbott, 2003).

Constraint- and score-based techniques are the most common and powerful methods to develop the BN. The former algorithm extracts dependencies within a large dataset when a few variables are strongly correlated. Multiple errors can be generated as a result of a single misleading independence test (Koller & Friedman, 2009). On the other hand, the latter technique looks at a structure learning problem as a general heuristic optimization approach by (a) hypothesizing the existence of a set of possible model solutions in the data, (b) computing scores for how well each model fits the data, (c) and choosing the model with the maximum score (Koller, 1998). In this research, we employed the score-based algorithm because it addresses scalability and data fitting and is more accurate and powerful when it comes to individual failure.

The two most common score functions are the Bayesian information criterion (BIC) and the maximum log-likelihood. While the maximum log-likelihood is usually suspected of overfitting of data, the BIC can overcome this shortcoming by introducing trade-offs between the likelihood of fitting the data and complexity of the model (Koller, 1998). The BIC can be computed from the following equation (Rissanen, 2007):

$$BIC = \sum_{i=1}^n \log P_{\mathbf{X}}(\mathbf{X}_i | \Pi \mathbf{X}_i) - \frac{d}{2} \log(n)$$

where d is the number of parameters in the global distribution, n is the number of samples, and the last term is introduced to reduce the effect of overfitting.

To obtain an optimal network and simulate annealing to locate high-scoring structures, this research employed two different BN learning score-based techniques: Tabu and hill climbing. These search algorithms have the advantages of obtaining a good solution with a minimum number of experiments when there is little prior knowledge of the problem. Moreover, these methods analyze all possible local changes in all steps and utilize the one that generates the largest improvement in the score (Sattari et al., 2020). HC is an iterative algorithm including a global search phase and a local search phase. The global search phase covers the search space as broadly as possible to determine a good initial solution for the local search phase. The local search then starts with this initial guess and employs a gradient-based technique to search around for a better option. As a result, the size of the search is reduced while the local search continues (Xi et al., 2004).

In short, HC starts with an initial guess of a solution, then iteratively makes local changes to find the best possible solution, but can get stuck in a local maximum (Jiang et al., 2013b). Some ways to avoid getting stuck include running many searches in parallel or probabilistically choosing the successor state. As there is no guarantee that HC can find the best answer (even though it often runs very quickly and gives good approximations), another search algorithm called Tabu search (TS) is applied. TS is a deterministic metaheuristic-based method presented separately by Hansen (1986) and Glover (2008). It uses a memory list to guide the search, starting with a feasible initial solution and picking the next best option that can increase the score function. In other words, after creating a list and evaluating a number of neighborhoods, the best one is accepted, even if it is low quality with respect to a cost function (Fig. 2). This search has the advantage of preventing a quick return to the same state (Jiang et al., 2013a).

All of the algorithms described above were implemented in MATLAB version 2018b and R version 3.5.1.

Table 1

Keywords used to identify PSM elements.

PSM element	Keyword
Compliance with Standards	compliance, comply, regulatory
Process Safety Information	psi
Hazard Identification & Risk Analysis	hazard, risk
Operating Procedures	procedure
Safe Work Practices	safety, safe work
Asset Integrity & Reliability	equipment, asset
Contractor Management	contractor, third party, 3rd party
Training	training, inexperience, lack of experience
Management Review & Continuous Improvement	management

3. Results & discussion

3.1. Incident data classification by keyword analysis

The first step was to remove incident reports that were missing root cause analysis. This resulted in a total of 16,473 incidents to include in the investigation: 7928 incident reports from 2014 and 8545 incident reports from 2015. Keyword analysis was used to determine the PSM element pertaining to each incident report. The relationship between PSM elements and keywords is shown in Table 1.

Every incident report that mentioned process safety also included the abbreviation, psi. As such, the keyword used to search for the *Process Safety Information* element was psi, and manual analysis was completed to remove incidents tagged due to irrelevant words (e.g., upside, collapsing, and the names of individuals).

From the 16,473 incidents analyzed, 8199 incidents violated at least one of the rules represented by the nine selected PSM elements (Table 2). Classifying the remaining incidents might be possible by completing a more detailed analysis or introducing more PSM elements into the analysis. The 8199 incidents were arranged by date to determine when these violations occurred (Appendix A).

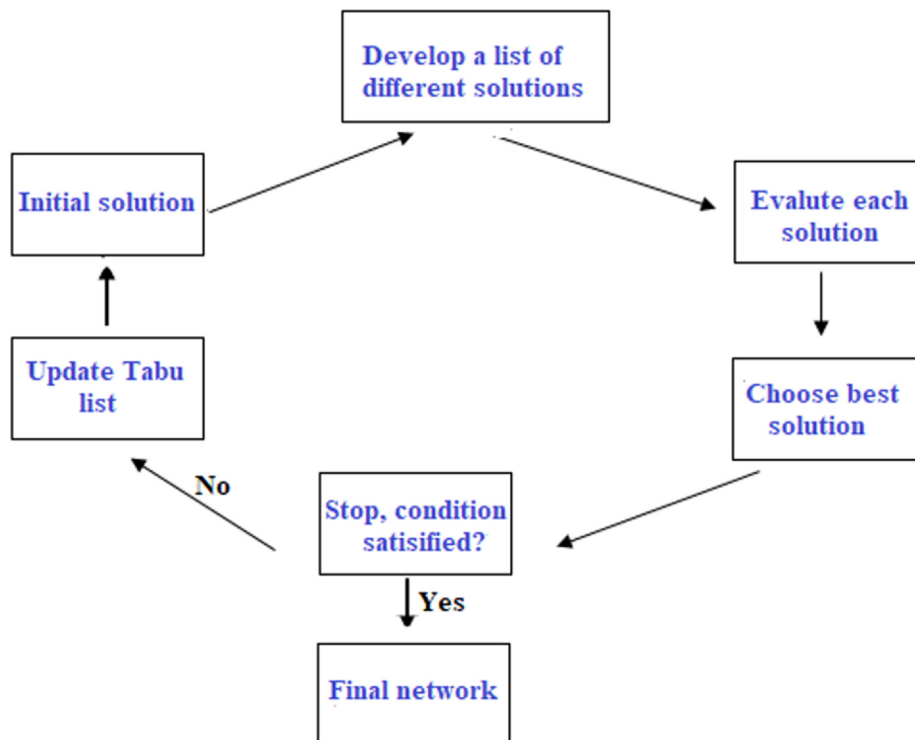


Fig. 2. Flowchart of a standard Tabu search algorithm.

Table 2
PSM elements or clusters.

Cluster	PSM Element
1	Compliance with Standards
2	Process Safety Information
3	Hazard Identification & Risk Analysis
4	Operating Procedures
5	Safe Work Practices
6	Asset Integrity & Reliability
7	Contractor Management
8	Training
9	Management Review & Continuous Improvement
10	Total Number of Incidents

3.2. Bayesian network

The previous cluster analysis step provided enough variables to develop the most probable reaction network (BN) representing the investigated network. Extracting information or knowledge from the large dataset undoubtedly requires the proper mechanism. Fundamentally, data mining applies different algorithms and methods to extract and discover patterns in stored data. BN, as a form of data mining, helped this study to derive useful, valid, unexpected, and understandable information from the large collection of data. The BN approach provided a graphical model that encodes probabilistic relationships between data, groups, or PSM elements, and was used to deal with uncertain knowledge in expert systems. After applying BNs, the two score-based search algorithms (Tabu & HC) generated the same results, providing certainty in the efficacy of the reaction network (Fig. 3).

The developed BN extracted very interesting information on how these PSM elements are not only related to the incident rate, but also their interconnectivity. For instance, while all nine PSM elements can be considered as the direct causes of incident rate, *Safe Work Practice* (group 5) or *Operating Procedures* (group 4) can also be the parents or causes of *Contractor Management* (group 7) and *Training* (group 8), respectively. It can also be inferred that lack of proper *Hazard Identification & Risk Analysis* (group 3) procedures can be, directly or indirectly, the cause of failure of many other PSM elements (groups 4, 5, 7, & 8).

To record the dependency and probable pathways between the clusters in a network, the strength value of all of the arcs were calculated (Table 3). The strongest dependency belongs to the highest

negative number and, in this case, the arc from group 6 (*Asset Integrity & Reliability*) to group 10 (*Total Number of Incidents*) shows the highest probability for updating itself (315.06). This implies that, once the arc is removed, the overall score of the network will automatically decrease by 315.06 and the *Total Number of Incidents* will be reduced by half. All around the world, companies have developed new techniques to meet the increasing demand for oil and gas. Consequently, the ongoing pressure to utilize aging facilities has resulted in more incidents and failures in plants. Unfortunately, between 1980 and 2006, 50% of major hazards and failures in oil and gas plants were related to the lack of poor asset integrity and reliability management systems (Ossai, 2012). Other research done by oil and gas producers confirms that major incidents can be reduced by focusing on asset integrity management (Hassan & Khan, 2012; Milanese et al., 2017).

The second highest dependency belongs to cluster 9 (*Management Review & Continuous Improvement*) and group 10 (*Total Number of Incidents*). In other words, organizations can decrease their incident rates by focusing on learning to learn (Bartnes et al., 2016) in addition to the commitment and support of management. This research focuses on providing recommendations on the elimination or reduction of these latent causes.

Another interesting finding extracted from this data is that, among the nine factors, group 1 (*Compliance with Standards*) has the least effect on the total incident rate (−3.49). This improvement in complying with standards could be the result of changes made by the Alberta Energy and Utilities Board (EUB) in 1994. Due to arguments related to budget reductions, the EUB shifted from a command-and-control regulatory system to a self-regulation system called “Compliance Assured by Audits and Penalty”. This new system has transferred responsibility to companies to make sure that their organizations must comply with the standards and guidelines. As a result, incident occurrences related to not following the standards have been reduced (Herremans & Welsh, 1999).

Process Safety Information (group 2) is the second least critical component impacting the total number of incidents (−4.73). For the past few decades, the gap between establishing safety information/metrics and incidents has increasingly been acknowledged and addressed in both Europe and the USA. As a result, many organizations such as the OSHA and CCPS have provided guidelines. The purpose of creating these standards is to reduce process safety incidents by focusing on creating and promoting safety metrics and safety information for organizations (Morrison et al., 2011).

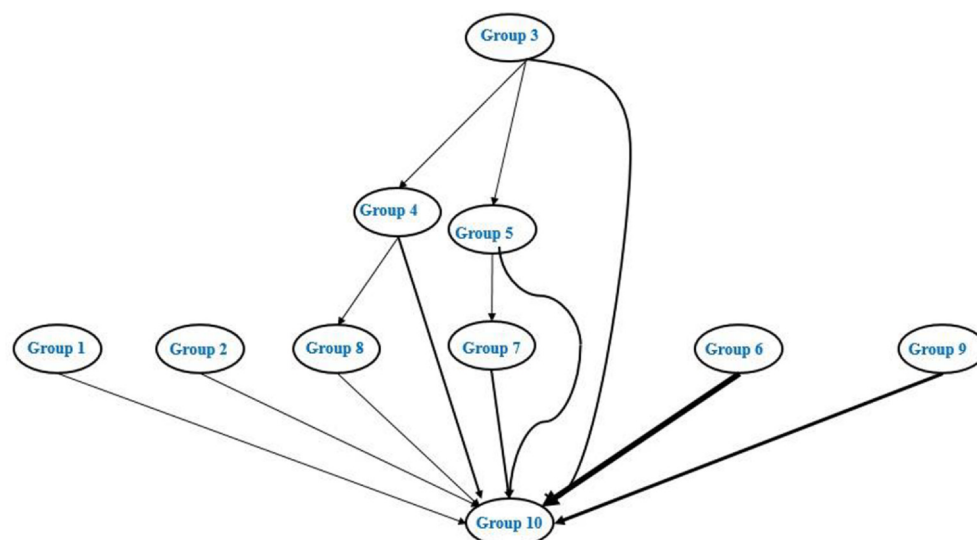


Fig. 3. Nine-cluster Bayesian network.

Table 3

Arc strength for the reaction network shown in Fig. 3 calculated by Tabu and HC.

From Cluster	To Cluster	Arc Strength
6 (Asset Integrity & Reliability)	10 (Total Number of Incidents)	-315.06
9 (Management Review & Continuous Improvement)	10 (Total Number of Incidents)	-102.35
3 (Hazard Identification & Risk Analysis)	10 (Total Number of Incidents)	-78.86
7 (Contractor Management)	10 (Total Number of Incidents)	-52.13
4 (Operating Procedures)	10 (Total Number of Incidents)	-50.86
5 (Safe Work Practices)	10 (Total Number of Incidents)	-30.81
8 (Training)	10 (Total Number of Incidents)	-9.96
2 (Process Safety Information)	10 (Total Number of Incidents)	-4.73
1 (Compliance with Standards)	10 (Total Number of Incidents)	-3.49
3 (Hazard Identification & Risk Analysis)	4 (Operating Procedures)	-7.84
3 (Hazard Identification & Risk Analysis)	5 (Safe Work Practices)	-4.63
4 (Operating Procedures)	8 (Training)	-1.67
5 (Safe Work Practices)	7 (Contractor Management)	-1.07

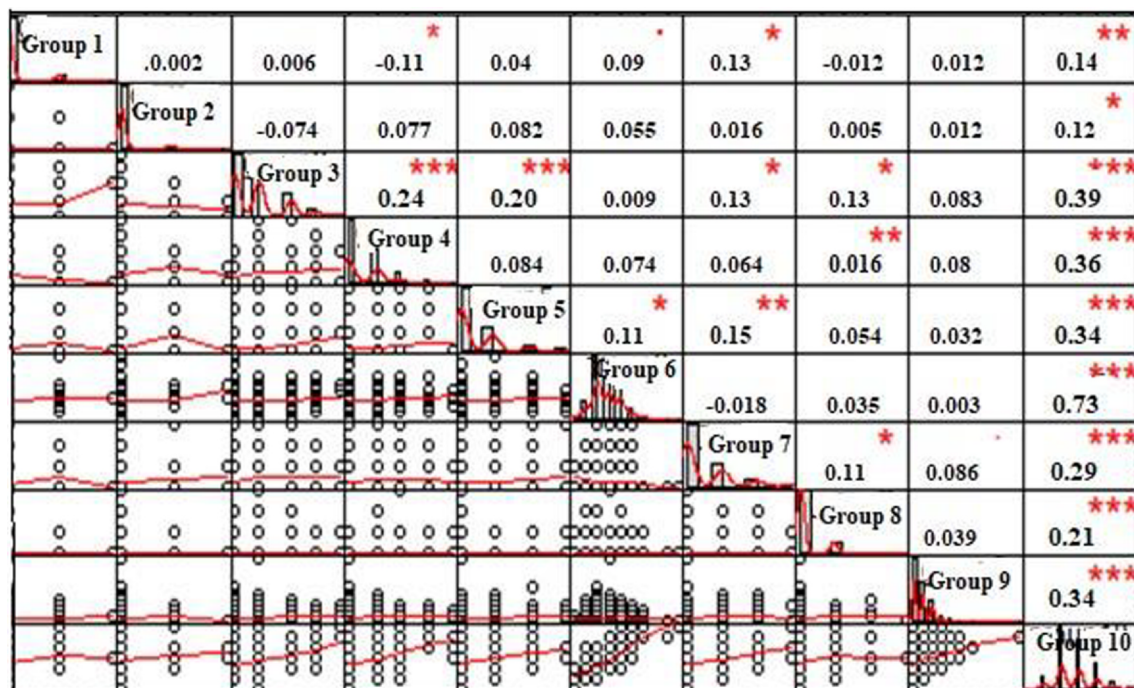


Fig. 4. Linear cross-correlation of the groups.

3.3. Cross correlation analysis

The map produced through the BN approach was validated using linear cross correlation of the clusters. Fig. 4 presents the results qualitatively and quantitatively in the lower and upper panels, respectively. The plot shows the distribution of each variable diagonally, the significance level by stars and the correlation values on the top of the diagonal, and the bivariate scatter plots with a fit line at the base of the diagonal.

The results are consistent with the findings in Table 3, with groups 10 and 6 having the most correlated variables (0.73) and groups 2 and 10 having the least (0.44). Overall, the correlation matrix shows a smooth correlation between variables, but that alone does not indicate causality among variables. Therefore, Bayesian learning was chosen to assess any reaction network between variables.

3.4. Discussion

We began by arguing that academic researchers must examine companies' data, use their PSM frameworks, and provide actionable recommendations to help improve their safety practices. We found that *Asset Integrity & Reliability* and *Management Review & Continuous*

Improvement are the most influential factors on the *Total Number of Incidents* for the oil and gas company studied. Thus, next we focus on providing recommendations for improving these elements of PSM. Regardless of the thoroughness of the PSM system, failure to successfully communicate maintenance activities, abnormal conditions, emergency response procedures, process hazards, and hundreds of other items of critical information can lead to disaster. Thus, recommendations will also be made with respect to maintaining successful communication interfaces between each involved employee or stakeholder, as well as promoting critical thinking and a safety culture within an organization.

3.4.1. Recommendations for reducing incident rate by focusing on asset integrity & reliability

Asset integrity requires that companies comply with all existing guidelines and standards, such as national, regulatory, and policies set by the company itself (Ramamany & Yusof, 2015). In an oil and gas industry, it is especially important for the company to manage assets through the management of their integrity. Asset integrity management (AIM) is used to create the framework for an organization to evaluate its processes and assets in connection with its objectives (Hassan & Khan, 2012). Asset integrity aims to look at an asset's ability to operate

accurately and efficiently while protecting other equipment and the well-being of personnel interacting with that asset. At the same time, maintaining risk and performance at a reasonable level is also a fundamental part of AIM (British Standards Institute, 2014). Focusing on asset integrity and working to improve AIM allows an industry to future-proof its investments. While protecting against losses, both minor and catastrophic, it also works to improve performance and profitability. Several challenges pertaining to asset integrity are faced by industries (Pirie & Østby, 2007; Dogan, 2014), such as:

- Poor or inadequate implementation of safe work practices;
- Operating in unpredictable circumstances, including environments with harsh weather;
- Improper documentation;
- Confusion caused by various non-standardized regulations published by different operator companies; and
- Poor collaborative efforts between companies operating within the same industry, including poor sharing of information and experiences.

By working to improve AIM, companies are hoping to improve regulatory requirements pertaining to health, safety, and the environment (HSE); documentation; access to information, including incident data; performance, of both assets and personnel; and risk management.

As technology continues to progress, fulfilling the goals to improve AIM also become more viable. Advancements in the field of computer science provide industries with highly efficient tools that increase control over complex systems. Such tools can be used to reduce human error and bias, properly organize and present information, and reduce risk. Information management systems (IMSS) are an example of such a tool that can be used to provide high quality asset integrity solutions (Goel et al., 2017). These can be developed to match management processes and ensure compliance with established regulations.

Another improvement could be the implementation of cloud-based software to better link the different parties involved within a project: technicians, manufacturers, regulators, end users, etc. Such interfaces can be used to provide diagnostics and increase communication between different stakeholders, a crucial aspect of AIM (Galar et al., 2012). Three steps can be used to describe the approach to improving AIM:

1. Creating a system to monitor the performance of assets and AIM systems.
2. Conducting focused studies on the most significant incidents observed and analyzing other areas of significant potential loss.
3. Developing systems to support AIM.

Moreover, many elements should be considered when developing a platform connecting asset integrity services and matching management activities. These components could include: concept and verification, data collection and compilation, technical and functional hierarchy, consequence classification analysis, material management and spare part evaluation, planning preventive maintenance (PM), performing PM, condition monitoring and measurement, reporting of maintenance and technical conditions, specific tools, analyzing and optimizing PM, integrity and barrier systems, and handover and training (Dogan, 2014). In addition, while most asset integrity management systems are considered after the project phase and for ageing assets, a system in which the life cycle assessment can be done during the design phase would be useful.

3.4.2. Recommendations for reducing incident rate by focusing on management review & continuous improvement

The Swiss Cheese model is a simple representation of incidents that do not occur due to a single problem but, instead, are due to the alignment of multiple deficiencies (Reason, 1990). Many incidents that

occur among frontline workers are the result of failures in design and choices made by management. As such, the commitment that management has towards promoting safety is a fundamental aspect of an organization's safety culture. Promoting safety among management can be accomplished by conducting management reviews to evaluate the performance of management and produce the desired results.

An effective method to improve performance is to learn from the past, such as through the analysis of historical events and data (Sonnemans & Körvers, 2006). If incidents are properly reported, these reports contain many instances of past shortcomings or failures that can be used as learning experiences to prevent similar incidents from re-occurring. Companies can use this knowledge to emphasize the fact that such incidents have already occurred, and thus are now foreseeable and preventable. Unfortunately, many incidents are not reported due to fear of legal responsibility in the case of managers or loss of occupation in the case of frontline workers (Davies, 2002). Being proactive, rather than reactive, can also play a big role in the prevention of an incident (e.g., paying special attention to the most frequent design or operational failures).

Managers are typically familiar with best practices in safety, but do not always act in accordance with the responsibilities expected of their position (Vaughan, 1996; Reason et al., 1998). This provides many areas for improvement: harmonizing safety practices and regular operating procedures, standardizing safety culture, increasing the competency of the workforce, and increasing the involvement of the workforce in decision-making. Moreover, managers need to motivate their employees to accept ownership of safety and the importance of near-miss reporting. However, the safety and stability of a process/operation requires the joint effort of management and supervision units, construction units, operation units, design units, and government (Lyu et al., 2019).

By focusing on continuous improvement, looking beyond the present and into the future becomes possible (Aven & Krohn, 2014). Continuous improvement requires constant attention and learning to improve every aspect of a business—production, engineering, and marketing—and involves relationships between individuals, work groups, and organizations. On a foundational level, influencing the actions, behaviors, and attitudes of individuals is simplest. Research on topics from psychology to management attempts to identify individual traits and behaviors that promote learning and creativity. Individuals learn and improve within organizations that support the open sharing of information (Garvin et al., 2008). Conversely, creativity declines within organizations that have high expectations towards conformity (Thompson, 2003). Organizational values that support risk-taking behavior and have tolerance for failure encourage individuals to learn and develop creativity.

3.4.3. Support and empower critical thinking, enhanced communication, and learning culture to improve organizational safety

Rules and regulations designed to promote safety within an organization are often feared due to their stringent nature (Reason et al., 1998). Rather than appearing strict, a robust safety management system should benefit organizational safety by providing workers at all levels within an organization with the knowledge, skills, and enthusiasm to learn, think critically, and communicate, specifically with regards to safety. In a study by Albert et al. (2014), the primary reason for hazards going unidentified was improper communication. More specifically, hazards were not identified because workers were unaware that such hazards were present on site as opposed to workers acting in an unsafe manner. Hence, this is an issue of organizational safety that can be improved when there is support and empowerment for workers to communicate, think critically, and intrinsically link knowledge.

Various types of knowledge are intrinsically linked, and these are influential in determining how rules and regulations are followed by workers. Within a professional setting, knowledge can be attained in the following three ways (Johannessen, 2006):

- Knowledge demonstrated by skill, which comes from day-to-day experience;
- Knowledge that is shared between employees or seniors; and
- Knowledge from educational settings and written sources.

Generally, workers at all levels within any high-risk industry, such as the oil and gas industry, are familiar with the knowledge and skills required to undertake tasks. These skills are more often than not acquired through internships, during training, or in an educational setting (Berglund, 2016). Hence, it can be implied that skills are an asset that can be shared between employees through dialogue; however, knowledge is an asset that becomes engraved in the mind through experience. Skills are unique to each employee based on their capabilities; however, they can be passed onto others in work environments that require collaboration and teamwork. This transmission of skills and knowledge within an organization is essential to improving the learning culture that thereby improves organizational safety. Learning culture can be defined as the established set of values, practices, and processes that promote continuous learning for its employees due to the belief that all systems significantly influence each other (Jin & Chen, 2013). Continuous learning requires critical thinking and communication. To build an effective learning culture, knowledge needs to be attained through objective analysis and evaluation of all aspects of the industry and then transmitted throughout the workforce.

In a high-risk industry, critical thinking is especially important and involves several processes. While it is difficult to provide a definitive system by which critical thinking occurs, the process generally requires the collection of information through observations and further application and analysis thereof (Hughes, 1996). To empower workers to think critically, these processes must involve succinct reasoning and reflection to form judgments and communicate them. When workers can undertake such processes, it leads them to be able to efficiently anticipate, detect, and manage risks and hazards.

One such method of empowering critical thinking is known as “kaizen,” the Japanese word for “improvement” (Hafey, 2017). While this concept has been around and in use by several industries, the term is rarely mentioned with regards to organizational safety. Kaizen was introduced by the Japanese and is most successfully used by their industries to promote continuous learning and improvement throughout entire organizations. The concept is simple: small positive changes lead to major improvements. Processes within an industry are improved by ensuring workers at all levels of an organization are involved, which empowers and increases the confidence of employees as they are supported by senior management (Berglund, 2020). The application of kaizen also promotes nonlinear learning, wherein knowledge is gained from being introduced to unfavorable events experienced by workers rather than observing workers during their daily routine procedures. This allows workers to become familiar with deviations and strengthens the ability to recognize both minor and major changes in processes. A method that promotes continuous learning engages workers in a process of critical thinking such that they are made aware of the presence of uncertainties in operations. This enables them to prepare for such situations, which eventually strengthens an organization’s safety culture.

To learn continuously, end solutions are learned but also evolving based on their relative effectiveness. The effectiveness of solutions to solve identified uncertainties can be measured and both qualitatively and quantitatively analyzed through data analytics, such as BN analysis. This flexibility creates a slow learning process wherein individuals can intervene constantly to find and solve problems (Sennett, 2008). The response by workers to a situation is dependent on their level of skill. Often, these skills are limited due to the linear nature of educational systems that are merely focused on providing technological knowledge. While knowledge is being transferred, it is not being thought about and hence the skills can at times be limited. This is highlighted by the fact that part-time or temporary workers make up

one-third of the labor force in most industries and half of the workers in the industry we studied, an increase from one-fifth of the labor force in previous years (Berglund, 2016). This reflects the stark decline in highly skilled worker labor force throughout various industries.

To empower critical thinking, a kaizen method coupled with an information technology (IT) system and training would ensure the transfer of knowledge rather than simply a skillset. Utilizing a system that involves data storage and machine learning methods to collect deviations in the process that reflect errors/hazards would enable trend analysis. This trend analysis would aid in developing updated process safety management policies to account for unrecognized hazards and further enhance the learning culture by improving workers’ attitudes towards and sufficiency in hazard recognition.

With the implementation of kaizen or continuous improvement comes room for building a dynamic risk management program. Oil and gas processes are complex in nature due to the numerous operations that are running simultaneously. In this risk management system, a Bayesian technique such as the one applied in this paper can be used to revise process parameters or inputs to the control system based on the outcomes of the assessment. A dynamic risk assessment that uses a Bayesian method would be useful as it provides the opportunity for kaizen while including deviations or uncertainties that are identified in the learning process and multiple variables (Khan et al., 2016).

Safety culture is an amalgamation of various factors that link learning culture, organizational safety, and critical thinking (Guldenmund, 2000). Communication plays a vital role in the way employees apply their critical thinking and further build a learning culture within an organization (Yeong & Shah Rollah, 2016). As discussed earlier, a robust safety culture ensures the safety management system adopted by an organization is successful when leaders and employees have the flexibility to communicate with utmost transparency. The values of an organization with respect to safety should be embedded in all members of an organization as a way of life. The burden of initiating the way to effective communication within an organization lies with the leaders, who must achieve several objectives to establish a positive safety culture (Louvar, 2013):

- Engaging with workers to strengthen professional relationships and create a positive safe work culture;
- Providing methods to improve communication skills to enable interdisciplinary transmission of knowledge;
- Organizing activities to relay messages throughout the organization to identify and prevent miscommunication while promoting values of the organization;
- Providing opportunities for employees to be involved in decision-making processes without the presence of decisions being made by the majority rather than individuals; this builds confidence and gives a positive image to the organization;
- Promoting bottom-to-top communication channels by allowing open conversations with the limitations of authority and cultural issues (as opposed to the typical top-to-bottom communication interface);
- Encouraging teamwork and providing training in the form of virtual reality games, online courses, or even training in a classroom setting to allow employees to interact and build relationships;
- Providing training to push the concept of working hard such that, in a situation that requires focus, all individuals can do so;
- Giving frequent opportunities for employees to believe in and relate to the organization’s vision, mission, and objectives by including them in daily routines; and
- Analyzing communication performance within an organization to assess areas of improvement by following the concept of continuous improvement.

Consideration of previous incidents at oil and gas facilities makes it evident that catastrophic disasters can occur when poor maintenance, irregular conditions, and process hazards (to name a few) are not

communicated (Berglund, 2020). Moreover, if emergency response procedures that are important when an incident takes place are not communicated, then limited knowledge is in place on actions that need to be taken to mitigate the consequences of a hazard. Without communication, even a well-thought-out PSM system does not benefit an organization. For this reason, it is of utmost importance to understand communication interfaces between employees and stakeholders who interact with each other. According to Wold & Laumann (2015), assessing the safety culture of an organization requires an evaluation of work procedures with regards to communication within an organization. Notably, a sense of class difference exists between superiors (executives) of an organization and workers performing tasks at the field level. As such, both groups have varying perceptions of PSM systems, which eventually impacts communication. This difference of mindset gives rise to misunderstanding as workers are unable to communicate freely and feel compelled to retain their opinions. To overcome these barriers in communication, it is important for workers to be introduced to an organization's PSM systems early on. While empowering critical thinking and communication is extremely important, psychological safety is equally responsible for employees' attitudes and behaviors towards such systems. Keeping these factors in mind, companies should use survey and interview data analysis to develop training programs, both in person and computer-based (such as games or through virtual reality), to showcase the importance of communication during critical scenarios. This training method can help workers foresee the benefits of following a PSM system and building a safety-first attitude.

For improved communications, the various groups and facilities existing in an organization must be identified (Mohammad et al., 2019), including internal and external stakeholders such as contractors. The number of separate groups within an organization determines the amount of time it will take for a communication to occur (Kelly & Berger, 2006). Even with the presence of guidelines, communicating effectively can be difficult in dynamic operating environments such as those in oil and gas facilities.

Interface management (IFM) is a system designed to improve communication by focusing on reducing the time it takes to send a message. IFM can be used to map various communication networks, including verbal and written communications between workers, contractors, and the general public (Kelly & Berger, 2006). An IFM system can be included in a general PSM system as it has not been used by oil and gas companies in their formal safety management systems. Through this IFM system, all members of the organization become aware of the communicational network. Moreover, IFM is flexible to account for normal operation and for emergency situations. Interaction between individuals and the system plays a significant role in process operations. Implementation of an IFM system provides a method to systematically monitor and control all communication channels for a particular process operation (Kelly & Berger, 2006). Most incidents that result from a lack of knowledge and fear of reporting can be avoided as IFM determines how communications are to be coordinated and who is responsible for passing on information.

Enhancing interfaces demonstrates that, despite one-on-one communication, there is also a possibility for bottom-to-top communication. An IFM system coupled with an IT-based safety management system allows workers to access the interface schematics related to their process operation when needed. Moreover, all workers can be made aware of each other's ongoing activities through an interactive IFM that is IT-based. The IFM framework will establish who is in charge and who is responsible for making decisions. This will vary depending on the scale of the organization. Large organizations will require several IFMs due to their complex communication network. Moreover, room for continuous improvement should be made as this is key to empowering critical thinking and a positive learning culture. Hence, upon creating an IFM framework and an IT-based PSM system, it is important to take feedback and intervene when appropriate to further enhance the safety culture of the organization. Workers must be involved in all stages to

boost confidence and positivity and to improve organizational safety.

4. Conclusion

Incident reporting systems play an important role in developing and maintaining safety procedures and efficient operations by detecting failures/incidents and providing lessons learned and recommendations before they become major accidents. The essence of this research was to develop a unique analytical approach by using AI and ML techniques to identify and prioritize the best strategies to reduce incident rates in an organization. To achieve this objective, this study used the elements of PSM defined by CCPS to classify 8199 incident reports from an oil and gas company. By utilizing a combination of root cause and keyword analysis, we were able to classify incident reports according to violations of the following nine PSM elements: *Compliance with Standards*, *Process Safety Information*, *Hazard Identification & Risk Analysis*, *Operating Procedures*, *Safe Work Practices*, *Asset Integrity & Reliability*, *Contractor Management*, *Training*, and *Management Review & Continuous Improvement*.

After cluster analysis, these variables were used as inputs for a Bayesian network model. Such a network can be used to determine relationships between the different PSM elements and the occurrence of incidents. This information would prove useful for a company as they determine where efforts should be focused to reduce the number of incidents occurring and minimize resulting losses. To identify an optimal structure for the Bayesian network, two different greedy search-and-score optimization algorithms were used: Tabu and hill climbing. Both methods resulted in the same network, thus providing confidence in its validity. More surprisingly, just two elements (*Asset Integrity & Reliability* and *Management Review & Continuous Improvement*) were found to be the latent cause for about 67% of incidents.

The methodology proposed in this research can be employed in many other fields such as reliability evaluation, quantitative risk assessment, diagnosis, prediction, automated reasoning, and developing a reaction network or model when there is a large dataset with limited prior knowledge of the system. This method provides the benefit of handling missing data while avoiding overfitting.

Furthermore, this research provides useful recommendations on improving and increasing the efficiency of these PSM elements. Developing a framework for asset integrity and reliability would be useful for ensuring a reduced incident rate. In complex industrial sectors, highly efficient IT tools can be utilized to achieve a desired functional working of assets during their lifetime, keeping them well maintained, controlling their impacts on health and environmental risks, as well as reducing human errors. Moreover, involving the workforce in safety activities, training, and decision-making, as well as harmonization and standardization of safety culture across the industry, are powerful approaches to develop an outstanding safety culture within an organization. Likewise, cultivating a safety management system that promotes critical thinking is especially important in a high-risk industry such as oil and gas, given that hazards may be unrecognized. Hence, being able to anticipate and evaluate such hazards is preferable and critical to the safety culture of an organization.

Further research from both computational and theoretical/practical points of view would be beneficial. The application of a hybrid structure of BN such as max-min hill climbing along with two greedy search-and-score algorithms (Tabu and HC) could be used to further maximize the optimal solution for developing the structure learning. It would also be beneficial to find a way to combine one qualitative approach—such as conducting interviews—with the completely quantitative approach taken in this research. Finally, developing more efficient algorithms that can generate the network and dependency between PSM elements within a limited computation time would be an asset.

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Appendix A. Supplementary material

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